Functional Requirement Specification (FRS)

Project: demo.opencart.com

Module: OpenCart Web Application

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1. Introduction

1.1 Purpose

The purpose of this document is to define the **functional requirements** of the OpenCart demo e-commerce website for performing structured testing activities including Functional, Integration, System, Regression, and UAT testing.

1.2 Scope

This FRS covers testing of:

- User Registration & Login
- Product Browsing
- Search Functionality
- Shopping Cart
- Checkout Process
- Payment
- Order Management
- Admin Panel (Basic overview)

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2. Overall Description

2.1 Product Overview

The system is an e-commerce web application built using the OpenCart platform that allows users to:

- Browse products
- Add products to cart
- Place orders
- Manage accounts
- Perform secure payments

2.2 User Roles

Role	Description
Guest User	Can browse and add to cart but limited access

Role	Description
Registered User	Can place orders and manage profile
Admin	Manages products, users, and orders

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3. Functional Requirements

3.1 Home Page

FR-HP-01

The system shall display:

- Logo
- Search bar
- Navigation menu
- Featured products
- Footer section

FR-HP-02

The system shall redirect users to product details when clicking a product.

3.2 User Registration

FR-REG-01

The system shall allow users to create an account using:

- First Name
- Last Name
- Email
- Telephone
- Password
- Confirm Password

FR-REG-02

The system shall validate:

- Mandatory fields
- Valid email format
- Password match

- Unique email ID

FR-REG-03

The system shall display success message upon successful registration.

3.3 User Login

FR-LOG-01

The system shall allow registered users to log in using valid credentials.

FR-LOG-02

The system shall display error message for:

- Invalid email
- Incorrect password

FR-LOG-03

The system shall allow logout functionality.

3.4 Product Browsing

FR-PROD-01

The system shall allow users to browse products by category.

FR-PROD-02

The system shall display product details:

- Product name
- Image
- Price
- Availability
- Description

FR-PROD-03

The system shall allow product sorting and filtering.

3.5 Search Functionality

FR-SRCH-01

The system shall allow users to search products using keywords.

FR-SRCH-02

The system shall display relevant search results.

FR-SRCH-03

The system shall display "No results found" for invalid search.

3.6 Shopping Cart

FR-CART-01

The system shall allow users to:

- Add product to cart
- Update quantity
- Remove product

FR-CART-02

The system shall calculate:

- Subtotal
- Taxes
- Total amount

FR-CART-03

The system shall persist cart items during session.

3.7 Checkout Process

FR-CHK-01

The system shall allow checkout for:

- Registered users
- Guest users

FR-CHK-02

The system shall collect:

- Billing address
- Shipping address
- Shipping method
- Payment method

FR-CHK-03

The system shall display order summary before confirmation.

3.8 Payment

FR-PAY-01

The system shall allow selection of available payment methods.

FR-PAY-02

The system shall process payment and confirm order.

FR-PAY-03

The system shall generate Order ID upon successful payment.

3.9 Order Management

FR-ORD-01

The system shall allow users to view order history.

FR-ORD-02

The system shall display:

- Order ID
- Date
- Status
- Total amount

3.10 Admin Panel (High-Level Testing Scope)

FR-ADM-01

The system shall allow admin login.

FR-ADM-02

The system shall allow admin to:

- Add/Edit/Delete products
- Manage categories
- View orders
- Manage customers

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4. Non-Functional Requirements (Testing Perspective)

4.1 Performance

- Page load time ≤ 3 seconds
- System shall support multiple concurrent users

4.2 Security

- Password must be encrypted
- Session timeout after inactivity
- Protection against SQL Injection and XSS

4.3 Usability

- Responsive design
- Proper error messages
- Easy navigation

4.4 Compatibility

- Chrome
- Firefox
- Edge
- Mobile browsers

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5. Assumptions

- Internet connectivity is stable.
- Payment gateway is functional.
- Test data is available.

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6. Constraints

- Demo site may reset data periodically.
- Limited payment options in demo version.

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7. Test Deliverables (Senior QA Level)

- Test Plan
- Test Scenarios
- Test Cases
- Test Data
- RTM (Requirement Traceability Matrix)
- Defect Reports
- Test Summary Report

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8. Conclusion

This FRS defines the complete functional behavior of the OpenCart demo e-commerce system to support comprehensive testing activities at industry level. It serves as a baseline document for test case creation, automation scripting, and defect validation.